Study notes

Metric Spaces & Topology

Chapters 3–5: Continuity, Completeness, Compactness

Tuesday 10 November 2:00–7:00 pm session

Definitions, theorems with proofs, worked examples, common mistakes, graded exercises with full solutions, proof templates, a counterexample bank and a glossary of every term needed for the exam.

How to use these notes

Each chapter starts with its **study slot** from the session plan below and a short list of what to focus on in that slot. Inside each chapter:

- **Intuition** paragraphs give the idea in plain English before the formal statement.
- **Definitions** are worded the way you should state them in an exam.
- **Theorems, lemmas, propositions** come with proofs of the results most often examined.
- **Examples** and **Warnings** cover standard cases and common mistakes.
- **Exercises** are tagged [Easy] , [Medium] or [Hard] and each has a full solution. Cover the solution and try the exercise first.

The appendices contain a **proof-template toolkit**, a **counterexample bank**, a **glossary** and the **recap checklist** for the 5:35 slot. Since these notes start at Chapter 3, Appendix A summarises everything from Chapters 0–2 that the later chapters use. Read it first if those chapters aren't fresh.

Chapter numbering follows the usual order for a first course in metric spaces. If your lecture notes order the topics differently, go by the section headings.

Session plan: Tuesday 10 November

Time	Task	Where
2:00–2:10	Warm-up: skim the chapter headings	Contents
2:10–2:30	Chapter 0: preliminaries	Appendix A
2:30–3:05	Chapter 1: metric spaces	Appendix A
3:05–3:15	Break	
3:15–3:50	Chapter 2: open and closed sets, topologies	Appendix A
3:50–4:25	Chapter 3: continuity	Chapter 3
4:25–4:40	Longer break	
4:40–5:10	Chapter 4: completeness	Chapter 4
5:10–5:35	Chapter 5: compactness	Chapter 5
5:35–5:45	Recap from memory	Recap checklist
5:45–6:00	Break	
6:00–6:50	Linear systems	Separate notes
6:50–7:00	Wrap-up: what's done, where to start next time	

Notation

(X, d)	a metric space; $x, y, z \in X$ are points
$B(x, r) = \{\, y \in X \;:\; d(x, y) < r \,\}$	open ball of radius $r > 0$
$\overline{B}(x, r) = \{\, y \in X \;:\; d(x, y) \leq r \,\}$	closed ball
$\operatorname{int} A,\ \operatorname{cl} A = \overline{A},\ \partial A$	interior, closure, boundary
$A^c = X \setminus A$	complement
A'	derived set (the limit points of A)
$(x_n)_{n\in\mathbb{N}},\ x_n \to x$	a sequence; convergence
$\mathbb{N} = \{1, 2, 3, \dots\},\ \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$	the usual number systems
$C[a, b]$	the continuous functions $[a, b] \to \mathbb{R}$

Contents

How to use these notes **i**

3 Continuity and Homeomorphisms **1**

3.1 Four equivalent definitions 1

3.2 Building continuous functions 2

3.3 Uniform continuity and Lipschitz maps 3

3.4 Homeomorphisms 3

3.5 Exercises 4

4 Completeness **6**

4.1 Cauchy sequences 6

4.2 Complete metric spaces 6

4.3 The Contraction Mapping Theorem 8

4.4 Further results 9

4.5 Exercises 9

5 Compactness **11**

5.1 Definitions 11

5.2 The main equivalence 12

5.3 Heine–Borel and its limits 12

5.4 Key theorems 13

5.5 Exercises 14

A Summary of Chapters 0–2 **16**

B Connectedness **17**

C Proof-Template Toolkit **18**

D Counterexample Bank **20**

E Glossary **21**

F Recap Checklist (5:35–5:45) **23**

Chapter 3

Continuity and Homeomorphisms

Study slot: 3:50–4:25 (35 min)

Learn the four equivalent definitions of continuity (Theorem 3.2) and when to use each. Memorise the hierarchy Lipschitz ⇒ uniformly continuous ⇒ continuous, with a counterexample for each converse. Be able to prove that two spaces are, or are not, homeomorphic.

Intuition. f is continuous if close inputs give close outputs. To make $f(y)$ within ε of $f(x)$, it is enough to take y within δ of x.

3.1 Four equivalent definitions

Definition 3.1 (ε–δ continuity)**.** Let $f\colon (X, d_X) \to (Y, d_Y)$. Then f is **continuous at** $a \in X$ if

$$\forall \varepsilon > 0 \;\; \exists \delta > 0 \;\; \forall x \in X : \quad d_X(x, a) < \delta \implies d_Y\big(f(x), f(a)\big) < \varepsilon.$$

Equivalently, $f(B(a,\delta)) \subseteq B(f(a),\varepsilon)$. We call f **continuous** if it is continuous at every point of X.

Theorem 3.2 (Characterisations of continuity)**.** *For $f\colon X \to Y$ between metric spaces, the following are equivalent:*

(1) f is continuous (Definition 3.1) at every point;

*(2) **(sequential)** $x_n \to x$ in X implies $f(x_n) \to f(x)$ in Y;*

*(3) **(open sets)** V open in Y implies $f^{-1}(V)$ open in X;*

*(4) **(closed sets)** C closed in Y implies $f^{-1}(C)$ closed in X.*

For topological spaces, (3) is taken as the definition *of continuity.*

Proof. **(1)⇒(3).** Let $V \subseteq Y$ be open and $a \in f^{-1}(V)$. Choose $\varepsilon > 0$ with $B(f(a),\varepsilon) \subseteq V$, and then δ as in (1). Then

$$f(B(a,\delta)) \subseteq B(f(a),\varepsilon) \subseteq V, \quad \text{so} \quad B(a,\delta) \subseteq f^{-1}(V).$$

(3)⇒(1). $B(f(a),\varepsilon)$ is open, so $f^{-1}(B(f(a),\varepsilon))$ is an open set containing a. It therefore contains some $B(a,\delta)$, which is exactly the statement in (1).

(3)⇔(4). Use $f^{-1}(Y \setminus C) = X \setminus f^{-1}(C)$ (Proposition 0.4).

(1)⇒(2). Given $\varepsilon > 0$, take δ from (1) and N with $d(x_n, x) < \delta$ for $n \geq N$. Then $d\big(f(x_n), f(x)\big) < \varepsilon$ for $n \geq N$.

(2)⇒(1) (contrapositive). Suppose (1) fails at a. Then there is $\varepsilon > 0$ such that for each $\delta = 1/n$ there is x_n with $d(x_n, a) < 1/n$ but $d\big(f(x_n), f(a)\big) \geq \varepsilon$. Thus $x_n \to a$ but $f(x_n) \not\to f(a)$. □

Exam tip (Which form to use).

- To show f *is* continuous: use ε–δ, or build f from known continuous maps (Proposition 3.3), or show f is Lipschitz.
- To show f is *not* continuous: give *one* sequence $x_n \to x$ with $f(x_n) \not\to f(x)$.
- To show a set is open or closed: write it as f^{-1}(open) or f^{-1}(closed) for a continuous f.
- In abstract or topological settings: use preimages of open sets.

3.2 Building continuous functions

Proposition 3.3.

(i) Constant maps and the identity $(X,d) \to (X,d)$ are continuous.

(ii) If $f\colon X \to Y$ and $g\colon Y \to Z$ are continuous, so is $g \circ f$, since $(g \circ f)^{-1}(W) = f^{-1}(g^{-1}(W))$.

(iii) If $f, g\colon X \to \mathbb{R}$ are continuous, so are $f \pm g$, fg, $|f|$, $\max\{f,g\}$, $\min\{f,g\}$, and f/g where $g \neq 0$.

(iv) $f = (f_1, \dots, f_n)\colon X \to \mathbb{R}^n$ is continuous $\iff$ each f_i is continuous.

(v) Every function from a discrete space is continuous.

(vi) For $\emptyset \neq A \subseteq X$, the map $x \mapsto d(x,A)$ is 1-Lipschitz: $|d(x,A) - d(y,A)| \leq d(x,y)$.

(vii) Restrictions of continuous maps to subspaces are continuous.

Example 3.4 (One-line proofs that sets are open or closed)**.**

- $\{\,(x,y) \in \mathbb{R}^2 \;:\; x^2 + y^2 < 1\,\} = f^{-1}((-\infty, 1))$ with $f(x,y) = x^2 + y^2$, so it is open.
- $\{\,(x,y) \in \mathbb{R}^2 \;:\; xy = 1\,\} = g^{-1}(\{1\})$ with $g(x,y) = xy$, so it is closed.
- $\left\{\, f \in C[0,1] \;:\; f(\tfrac{1}{2}) > 0 \,\right\}$ is open in d_∞, because evaluation $E(f) = f(\frac{1}{2})$ satisfies $|E(f) - E(g)| \leq d_\infty(f,g)$.
- The zero set $\{\,x \;:\; f(x) = 0\,\}$ of a continuous $f\colon X \to \mathbb{R}$ is closed.

Warning (Images are badly behaved). Continuous maps need not send open sets to open sets, or closed sets to closed sets:

$$x \mapsto x^2 :\ (-1,1) \longmapsto [0,1), \qquad \arctan :\ \mathbb{R} \longmapsto (-\tfrac{\pi}{2}, \tfrac{\pi}{2}).$$

Projecting the closed hyperbola $\{xy = 1\}$ to the x-axis gives $\mathbb{R} \setminus \{0\}$, which is not closed. Only *preimages* behave well, apart from compact sets (Theorem 5.7).

Warning (Continuity depends on the metrics). $\mathrm{id}\colon (\mathbb{R}, \text{discrete}) \to (\mathbb{R}, |\cdot|)$ is continuous, but $\mathrm{id}\colon (\mathbb{R}, |\cdot|) \to (\mathbb{R}, \text{discrete})$ is not: $\mathrm{id}^{-1}(\{0\}) = \{0\}$ is not open. Likewise $\mathrm{id}\colon (C[0,1], d_\infty) \to (C[0,1], d_1)$ is continuous, since $d_1 \leq d_\infty$, but its inverse is not (Example 1.17).

3.3 Uniform continuity and Lipschitz maps

Definition 3.5. Let $f\colon (X, d_X) \to (Y, d_Y)$.

(i) f is **uniformly continuous** if

$$\forall \varepsilon > 0 \;\; \exists \delta > 0 \;\; \forall x, y \in X : \quad d_X(x,y) < \delta \implies d_Y\big(f(x), f(y)\big) < \varepsilon.$$

Here δ depends only on ε, not on the point. Compare the position of "$\forall x$" with Definition 3.1.

(ii) f is **Lipschitz** with constant $L \geq 0$ if $d_Y\big(f(x), f(y)\big) \leq L\, d_X(x,y)$ for all $x, y \in X$. If $L < 1$, f is a **contraction** (Chapter 4).

Proposition 3.6. *Lipschitz $\Rightarrow$ uniformly continuous $\Rightarrow$ continuous. Neither implication reverses.*

Proof. If f is L-Lipschitz with $L > 0$, take $\delta = \varepsilon / L$. The second implication is immediate from the definitions. The counterexamples are in the table below. $\square$

Function	Continuous	Uniformly continuous	Lipschitz
x^2 on $\mathbb{R}$	✓	✗	✗
$1/x$ on $(0,1)$	✓	✗	✗
$\sin(1/x)$ on $(0,1)$	✓	✗	✗
$\sqrt{x}$ on $[0,1]$	✓	✓ (compact domain)	✗ (slope $\to \infty$ at 0)
$\sin x$ on $\mathbb{R}$	✓	✓	✓ ($L = 1$)
x^2 on $[0,1]$	✓	✓	✓ ($L = 2$)

Proposition 3.7 (Sequential criterion)**.** *f is* not *uniformly continuous $\iff$ there exist $\varepsilon > 0$ and sequences $(x_n), (y_n)$ with*

$$d(x_n, y_n) \to 0 \quad \text{but} \quad d\big(f(x_n), f(y_n)\big) \geq \varepsilon \;\; \text{for all } n.$$

Example 3.8 (x^2 is not uniformly continuous on $\mathbb{R}$)**.** Take $x_n = n$ and $y_n = n + \frac{1}{n}$. Then $|x_n - y_n| = \frac{1}{n} \to 0$, but

$$y_n^2 - x_n^2 = 2 + \frac{1}{n^2} \geq 2.$$

Proposition 3.9. *A uniformly continuous map sends Cauchy sequences to Cauchy sequences. A continuous map need not: $1/x$ on $(0,1)$ sends the Cauchy sequence $(1/n)$ to (n).*

3.4 Homeomorphisms

Definition 3.10. A **homeomorphism** is a bijection $f\colon X \to Y$ such that both f and f^{-1} are continuous. If one exists, X and Y are **homeomorphic**, written $X \cong Y$. A **topological property** is one preserved by homeomorphisms.

Intuition. A homeomorphism stretches and bends without tearing or gluing. This is why topologists treat a coffee mug and a doughnut as the same space.

Example 3.11.

(a) $(-1,1) \cong \mathbb{R}$ via $x \mapsto \dfrac{x}{1-|x|}$, with inverse $y \mapsto \dfrac{y}{1+|y|}$. Alternatively, use $x \mapsto \tan(\pi x/2)$.

(b) Any two open intervals are homeomorphic, and so are any two closed intervals (use affine maps $x \mapsto \alpha x + \beta$).

(c) An isometry is a homeomorphism onto its image.

(d) If d, d' are topologically equivalent, then $\mathrm{id}\colon (X,d) \to (X,d')$ is a homeomorphism.

Warning (A continuous bijection need not be a homeomorphism). The map $f\colon [0,2\pi) \to S^1$, $f(t) = (\cos t, \sin t)$, is a continuous bijection, but f^{-1} is not continuous at $(1,0)$. Points just below $(1,0)$ on the circle are sent to numbers near 2π, not near 0. Another example is $\mathrm{id}\colon (\mathbb{R}, \text{discrete}) \to (\mathbb{R}, |\cdot|)$. A continuous bijection *is* a homeomorphism when the domain is compact and the codomain is Hausdorff (Theorem 5.10).

Topological (preserved by $\cong$)	**Not topological** (depends on the metric)
open and closed sets, convergence, continuity, Hausdorff, compactness, connectedness, separability, number of isolated points	boundedness ($\mathbb{R} \cong (-1,1)$), completeness ($\mathbb{R} \cong (0,1)$), being Cauchy, uniform continuity, diameter

Exam tip (Proving $X \not\cong Y$). Find a topological property that one space has and the other lacks. $[0,1] \not\cong (0,1)$ because only $[0,1]$ is compact. $\mathbb{R} \not\cong \mathbb{R}^2$ because removing a point disconnects $\mathbb{R}$ but not $\mathbb{R}^2$. $[0,1) \not\cong (0,1)$ because removing 0 leaves $[0,1)$ connected, while removing any point disconnects $(0,1)$.

3.5 Exercises

Exercise 3.1. [Easy] Prove from the definition that $f(x) = 3x + 2$ is continuous on $\mathbb{R}$.

Solution. Given $\varepsilon > 0$, put $\delta = \varepsilon/3$. If $|x-a| < \delta$, then $|f(x) - f(a)| = 3\,|x-a| < \varepsilon$. In fact f is 3-Lipschitz, so it is uniformly continuous. □

Exercise 3.2. [Easy] Let $f = \mathbf{1}_{[0,\infty)}$, that is, $f(x) = 1$ for $x \geq 0$ and $f(x) = 0$ for $x < 0$. Show f is not continuous in two ways.

Solution. Sequences: $x_n = -\frac{1}{n} \to 0$, but $f(x_n) = 0 \not\to 1 = f(0)$. *Open sets:* $f^{-1}((\tfrac{1}{2}, 2)) = [0,\infty)$ is not open. □

Exercise 3.3. [Medium] For $\emptyset \neq A \subseteq X$, prove $|d(x,A) - d(y,A)| \leq d(x,y)$. Deduce that $\operatorname{cl} A = \{\, x \,:\, d(x,A) = 0 \,\}$ is closed.

Solution. For every $a \in A$, $d(x,A) \leq d(x,a) \leq d(x,y) + d(y,a)$. Taking the infimum over a gives $d(x,A) \leq d(x,y) + d(y,A)$. Swapping x and y gives the reverse inequality. So $x \mapsto d(x,A)$ is continuous, and $\operatorname{cl} A$ is the preimage of the closed set $\{0\}$ (Theorem 2.9). □

Exercise 3.4. [Medium] Show that $I\colon C[0,1] \to \mathbb{R}$, $I(f) = \int_0^1 f$, is continuous for both d_∞ and d_1.

Solution. $|I(f) - I(g)| = \left|\int_0^1 (f-g)\right| \le \int_0^1 |f-g| = d_1(f,g) \le d_\infty(f,g)$, so I is 1-Lipschitz for both metrics. □

Exercise 3.5. [Hard] Is $E(f) = f(0)$ continuous from $(C[0,1], d_1)$ to $\mathbb{R}$?

Solution. No. Let $f_n(t) = \max\{0, 1-nt\}$, a spike of height 1 at 0. Then $d_1(f_n, 0) = \frac{1}{2n} \to 0$, so $f_n \to 0$. But $E(f_n) = 1 \not\to 0 = E(0)$. (With d_∞, E is 1-Lipschitz.) □

Exercise 3.6. [Hard] Let $f, g\colon X \to Y$ be continuous, where Y is a metric space. Show $\{\, x \,:\, f(x) = g(x) \,\}$ is closed. Deduce that if $f = g$ on a dense set, then $f = g$.

Solution. $h(x) = d_Y\big(f(x), g(x)\big)$ is continuous, since $x_n \to x$ implies $f(x_n) \to f(x)$ and $g(x_n) \to g(x)$, and then $h(x_n) \to h(x)$ by Lemma 1.12(iii). The set is $h^{-1}(\{0\})$, which is closed. If it contains a dense D, it contains $\operatorname{cl} D = X$. For example, a continuous function on $\mathbb{R}$ is determined by its values on $\mathbb{Q}$. □

Chapter 4

Completeness

Study slot: 4:40–5:10 (30 min)

Learn the definition of a Cauchy sequence, the list of complete and incomplete spaces, Theorem 4.6 (closed $\iff$ complete), and the full proof of the Contraction Mapping Theorem (Theorem 4.9), a favourite long exam question.

Intuition. A sequence is Cauchy if its terms eventually get as close to *each other* as you like, so they look as if they are converging. Completeness says that whenever this happens, there really is a limit in the space. $\mathbb{Q}$ is incomplete because $1,\ 1.4,\ 1.41,\ 1.414, \ldots$ "wants" to converge to $\sqrt{2} \notin \mathbb{Q}$. The space has holes.

4.1 Cauchy sequences

Definition 4.1 (Cauchy sequence)**.** A sequence $(x_n)_{n\in\mathbb{N}}$ in (X, d) is **Cauchy** if

$$\forall \varepsilon > 0 \ \ \exists N \in \mathbb{N} \ \ \forall m, n \geq N : \quad d(x_m, x_n) < \varepsilon.$$

Unlike the definition of convergence, this does not mention a limit.

Proposition 4.2.

(i) Every convergent sequence is Cauchy.

(ii) Every Cauchy sequence is bounded.

(iii) A Cauchy sequence with a convergent subsequence converges, to the same limit.

Proof. (i) If $x_n \to x$, choose N with $d(x_n, x) < \varepsilon/2$ for $n \geq N$. Then for $m, n \geq N$,

$$d(x_m, x_n) \leq d(x_m, x) + d(x, x_n) < \varepsilon.$$

(ii) With $\varepsilon = 1$, all x_n with $n \geq N$ lie in $B(x_N, 1)$. So every term lies in $B(x_N, R)$ with $R = 1 + \max_{i<N} d(x_i, x_N)$. (iii) Let $x_{n_k} \to x$ and $\varepsilon > 0$. Choose N with $d(x_m, x_n) < \varepsilon/2$ for $m, n \geq N$, then k with $n_k \geq N$ and $d(x_{n_k}, x) < \varepsilon/2$. For $n \geq N$,

$$d(x_n, x) \leq d(x_n, x_{n_k}) + d(x_{n_k}, x) < \varepsilon. \qquad \square$$

4.2 Complete metric spaces

Definition 4.3 (Complete space)**.** (X, d) is **complete** if every Cauchy sequence in X converges to a point *of* X. A complete normed vector space is a **Banach space**.

Example 4.4 (Complete spaces)**.**

(a) $\mathbb{R}$. A Cauchy sequence is bounded, so it has a convergent subsequence by Bolzano–Weierstrass, so it converges by Proposition 4.2(iii). This uses the completeness axiom.

(b) $(\mathbb{R}^n, d_p)$ for $p = 1, 2, \infty$: each coordinate sequence is Cauchy in $\mathbb{R}$.

(c) Every closed subset of $\mathbb{R}^n$, such as $[0,1]$, $\mathbb{Z}$ or the Cantor set (Theorem 4.6).

(d) Every discrete metric space: with $\varepsilon = \frac{1}{2}$, a Cauchy sequence is eventually constant.

(e) $(C[a,b], d_\infty)$ (Theorem 4.7).

(f) $\ell^1, \ell^2, \ell^\infty$, and every compact metric space (Chapter 5).

Example 4.5 (Incomplete spaces and the Cauchy sequence that shows it)**.**

Space	Cauchy sequence with no limit in the space
$\mathbb{Q}$	decimal truncations of $\sqrt{2}$: $1,\ 1.4,\ 1.41, \cdots \to \sqrt{2} \notin \mathbb{Q}$
$(0,1)$	$x_n = \frac{1}{n} \to 0 \notin (0,1)$
$\mathbb{R} \setminus \{0\}$	$x_n = \frac{1}{n}$
$(C[0,1], d_1)$	continuous ramps tending to a step function (Exercise 4.4)
polynomials on $[0,1]$ with d_∞	Taylor partial sums $\sum_{k \le n} t^k/k! \to e^t$, which is not a polynomial
$\mathbb{R}$ with $d(x,y) = \lvert\arctan x - \arctan y\rvert$	$x_n = n$, since $\arctan n \to \frac{\pi}{2}$

Warning (Completeness is not a topological property)*.* $\mathbb{R} \cong (0,1)$, but only $\mathbb{R}$ is complete. The last row above gives $\mathbb{R}$ with its usual topology but an incomplete metric. Completeness *is* preserved by Lipschitz equivalent metrics, since they have the same Cauchy sequences.

Theorem 4.6 (Closed $\iff$ complete)**.**

(i) If X is complete and $F \subseteq X$ is closed, then F is complete.

(ii) If $Y \subseteq X$ is complete (with the induced metric), then Y is closed in X. This holds in any *metric space X.*

In particular, a subset of a complete metric space is complete if and only if it is closed.

Proof. (i) A Cauchy sequence in F is Cauchy in X, so it converges to some $x \in X$. Since F is closed, $x \in F$ by Theorem 2.14(iv). (ii) Let $y_n \in Y$ with $y_n \to x \in X$. Then (y_n) is convergent, hence Cauchy, so $y_n \to y$ for some $y \in Y$. By uniqueness of limits, $x = y \in Y$. □

Theorem 4.7. *$(C[a,b], d_\infty)$ is complete.*

Proof. Let (f_n) be Cauchy for d_∞.

Step 1 (candidate limit). For each $t \in [a,b]$, $|f_n(t) - f_m(t)| \le d_\infty(f_n, f_m)$, so $(f_n(t))$ is Cauchy in $\mathbb{R}$ and converges. Define $f(t) = \lim_n f_n(t)$.

Step 2 (convergence in the metric). Given $\varepsilon > 0$, choose N with $|f_n(t) - f_m(t)| < \varepsilon$ for all t and all $m, n \ge N$. Letting $m \to \infty$ gives $|f_n(t) - f(t)| \le \varepsilon$ for all t and $n \ge N$, that is, $\sup_t |f_n(t) - f(t)| \to 0$.

Step 3 (the limit lies in the space; the $\varepsilon/3$ argument). Fix n with $\sup_t |f_n(t) - f(t)| < \varepsilon/3$, and

$\delta > 0$ with $|f_n(t) - f_n(s)| < \varepsilon/3$ whenever $|t - s| < \delta$. Then for $|t - s| < \delta$,

$$|f(t) - f(s)| \le |f(t) - f_n(t)| + |f_n(t) - f_n(s)| + |f_n(s) - f(s)| < \varepsilon.$$

So $f \in C[a, b]$ and $d_\infty(f_n, f) \to 0$. □

Exam tip (Template for "prove X is complete"). (1) Take a Cauchy sequence and find a candidate limit, usually pointwise or coordinatewise. (2) Show convergence to the candidate *in the metric*. (3) Show the candidate belongs to X.

4.3 The Contraction Mapping Theorem

Definition 4.8. A map $f\colon X \to X$ is a **contraction** if there is a constant $k \in [0, 1)$ with

$$d(f(x), f(y)) \le k\, d(x, y) \qquad \text{for all } x, y \in X.$$

A **fixed point** of f is a point x^* with $f(x^*) = x^*$.

Theorem 4.9 (Contraction Mapping Theorem / Banach Fixed Point Theorem)**.** *Let (X, d) be a* non-empty complete *metric space and $f\colon X \to X$ a contraction with constant k. Then:*

(i) f has a unique *fixed point x^*;*

(ii) for every $x_0 \in X$, the iterates $x_{n+1} = f(x_n)$ converge to x^*;

(iii) (a priori error bound) $d(x_n, x^*) \le \dfrac{k^n}{1 - k}\, d(x_0, x_1)$.

Proof. Step 1. By induction, $d(x_n, x_{n+1}) = d(f(x_{n-1}), f(x_n)) \le k\, d(x_{n-1}, x_n) \le \cdots \le k^n d(x_0, x_1)$.

Step 2 (Cauchy). For $m > n$, by the polygon inequality and the geometric series,

$$d(x_n, x_m) \le \sum_{j=n}^{m-1} d(x_j, x_{j+1}) \le d(x_0, x_1) \sum_{j=n}^{\infty} k^j = \frac{k^n}{1 - k}\, d(x_0, x_1) \xrightarrow[n\to\infty]{} 0,$$

because $0 \le k < 1$.

Step 3 (convergence). X is complete, so $x_n \to x^*$ for some $x^* \in X$.

Step 4 (fixed point). f is Lipschitz, hence continuous, so

$$f(x^*) = f\big(\lim_n x_n\big) = \lim_n f(x_n) = \lim_n x_{n+1} = x^*.$$

Step 5 (uniqueness). If $f(x) = x$ and $f(y) = y$, then $d(x, y) = d(f(x), f(y)) \le k\, d(x, y)$. So $(1 - k)\, d(x, y) \le 0$, giving $d(x, y) = 0$.

Finally, (iii) follows by letting $m \to \infty$ in Step 2, using continuity of d. □

Warning (Every hypothesis is needed).

- *Completeness:* $f(x) = x/2$ on $(0, 1]$ is a contraction with no fixed point.
- *A uniform $k < 1$:* $f(x) = x + \frac{1}{x}$ on $[1, \infty)$ satisfies $|f(x) - f(y)| = |x - y|\,(1 - \frac{1}{xy}) < |x - y|$ for $x \ne y$, but $f(x) = x$ has no solution.
- *$k = 1$ is not enough:* $f(x) = x + 1$ on $\mathbb{R}$.

- $f(X) \subseteq X$*:* always check this first in applications.

Example 4.10 (Solving $\cos x = x$)**.** Take $X = [0,1]$, which is closed in $\mathbb{R}$ and hence complete, and $f = \cos$. Then:

(i) $f(X) = [\cos 1, 1] \subseteq [0,1]$;

(ii) by the Mean Value Theorem, $|\cos x - \cos y| = |\sin \xi|\,|x-y| \le \sin(1)\,|x-y|$, and $k = \sin 1 \approx 0.84 < 1$.

Therefore $\cos x = x$ has exactly one solution in $[0,1]$ (namely $x^* \approx 0.739$), and iterating cos from any starting point converges to it.

Remark 4.11 (Further applications). The Picard–Lindelöf theorem on existence and uniqueness for $y' = F(t,y)$ applies Theorem 4.9 to the integral operator

$$(Ty)(t) = y_0 + \int_{t_0}^{t} F(s, y(s))\, ds$$

on a closed subset of $(C[a,b], d_\infty)$. Other applications include integral equations, the convergence of Newton's method, and the inverse function theorem.

Exam tip (Recipe for applying the theorem). (1) Name a complete space X and justify it. (2) Check $f(X) \subseteq X$. (3) Find $k < 1$, typically using $\sup |f'| < 1$ and the Mean Value Theorem. (4) Conclude there is a unique fixed point, which is the limit of the iterates.

4.4 Further results

Theorem 4.12 (Cantor Intersection Theorem)**.** *Let X be complete and $F_1 \supseteq F_2 \supseteq \cdots$ non-empty closed subsets with $\operatorname{diam} F_n \to 0$. Then $\bigcap_{n=1}^{\infty} F_n$ consists of exactly one point.*

Proof. Pick $x_n \in F_n$. For $m \ge n$, both x_m and x_n lie in F_n, so $d(x_m, x_n) \le \operatorname{diam} F_n \to 0$. Thus (x_n) is Cauchy and converges to some x. Each F_n is closed and contains the tail $(x_m)_{m \ge n}$, so $x \in F_n$ for every n. If y also lies in the intersection, then $d(x,y) \le \operatorname{diam} F_n \to 0$, so $y = x$. □

The hypotheses are needed: $\bigcap_n (0, \frac{1}{n}) = \emptyset$ (not closed), and $\bigcap_n [n, \infty) = \emptyset$ (the diameters do not tend to 0).

Theorem 4.13 (Baire Category Theorem)**.** *In a complete metric space, every countable intersection of dense open sets is dense. Equivalently, a non-empty complete metric space is not a countable union of nowhere dense sets.*

Consequences: $\mathbb{R}$ is uncountable; $\mathbb{Q}$ is not complete under any metric topologically equivalent to the usual one; continuous nowhere-differentiable functions exist.

Theorem 4.14 (Completion)**.** *Every metric space X embeds isometrically as a dense subset of a complete metric space $\widehat{X}$, which is unique up to isometry. For example, $\widehat{\mathbb{Q}} = \mathbb{R}$.*

4.5 Exercises

Exercise 4.1. [Easy] Are $[0,\infty)$, $\mathbb{Z}$ and $\left\{ \frac{1}{n} \,:\, n \in \mathbb{N} \right\}$ complete?

Solution. $[0,\infty)$: yes, it is closed in $\mathbb{R}$. $\mathbb{Z}$: yes, it is closed, and its Cauchy sequences are eventually constant. $\left\{\frac{1}{n} : n \in \mathbb{N}\right\}$: no, because $(\frac{1}{n})$ is Cauchy with limit 0 outside the set. □

Exercise 4.2. [Easy] Show that $x_n = \sum_{k=1}^{n} \frac{1}{k}$ is not Cauchy.

Solution. $x_{2n} - x_n = \displaystyle\sum_{k=n+1}^{2n} \frac{1}{k} \geq n \cdot \frac{1}{2n} = \frac{1}{2}$ for every n, so no N works for $\varepsilon = \frac{1}{2}$. Note that $x_{n+1} - x_n \to 0$ is *not* enough for a sequence to be Cauchy. □

Exercise 4.3. [Medium] Show that $f(x) = \frac{1}{2}(x + \frac{2}{x})$ is a contraction of $[1,2]$ and find its fixed point.

Solution. $f'(x) = \frac{1}{2}(1 - \frac{2}{x^2}) \in [-\frac{1}{2}, \frac{1}{4}]$ on $[1,2]$, so $|f'| \leq \frac{1}{2}$ and f is a contraction with $k = \frac{1}{2}$. Also f has minimum $f(\sqrt{2}) = \sqrt{2}$ and $f(1) = f(2) = \frac{3}{2}$, so $f([1,2]) \subseteq [\sqrt{2}, \frac{3}{2}] \subseteq [1,2]$. The fixed point solves $x = \frac{1}{2}(x + \frac{2}{x})$, so $x^2 = 2$ and $x^* = \sqrt{2}$. On $[1,2] \cap \mathbb{Q}$, f is still a contraction of the space into itself but has no fixed point, which shows completeness is needed. □

Exercise 4.4. [Hard] Show that $(C[0,1], d_1)$ is not complete.

Solution. For $n \geq 2$ let

$$f_n(t) = \begin{cases} 0, & 0 \leq t \leq \frac{1}{2}, \\ n(t - \frac{1}{2}), & \frac{1}{2} \leq t \leq \frac{1}{2} + \frac{1}{n}, \\ 1, & \frac{1}{2} + \frac{1}{n} \leq t \leq 1. \end{cases}$$

For $m > n$, f_m and f_n differ only on $[\frac{1}{2}, \frac{1}{2} + \frac{1}{n}]$, and by at most 1, so $d_1(f_m, f_n) \leq \frac{1}{n}$. The sequence is Cauchy.

Suppose $d_1(f_n, f) \to 0$ with $f \in C[0,1]$. Then $\int_0^{1/2} |f| = \int_0^{1/2} |f - f_n| \leq d_1(f, f_n) \to 0$, so $f = 0$ on $[0, \frac{1}{2}]$ by continuity. For any $c > \frac{1}{2}$ and all n with $\frac{1}{2} + \frac{1}{n} < c$, $\int_c^1 |f - 1| \leq d_1(f, f_n) \to 0$, so $f = 1$ on $[c, 1]$. Hence $f = 1$ on $(\frac{1}{2}, 1]$. Then f is discontinuous at $\frac{1}{2}$, a contradiction. □

Exercise 4.5. [Hard] Let X be complete and $f\colon X \to X$ such that the iterate f^m is a contraction for some $m \geq 1$. Prove that f has a unique fixed point.

Solution. By Theorem 4.9, f^m has a unique fixed point x^*. Then $f^m(f(x^*)) = f(f^m(x^*)) = f(x^*)$, so $f(x^*)$ is also a fixed point of f^m. By uniqueness, $f(x^*) = x^*$. Any fixed point of f is a fixed point of f^m, so it is unique. □

Chapter 5

Compactness

Study slot: 5:10–5:35 (25 min)

Learn the open-cover definition word for word, the three equivalent forms for metric spaces (Theorem 5.3), Heine–Borel and where it fails, and the theorems about continuous maps on compact spaces. Practise proving that a set is *not* compact by giving a bad open cover or a bad sequence.

Intuition. Compactness is the topological version of "finite". Finite sets have convenient properties: every real function on them has a maximum, and every sequence in them repeats some value infinitely often. Compact sets keep these properties even when infinite. In $\mathbb{R}^n$, compact simply means *closed and bounded*.

5.1 Definitions

Definition 5.1 (Open cover, compactness)**.** Let $K \subseteq X$.

(i) An **open cover** of K is a family $\mathcal{U} = \{U_i\}_{i \in I}$ of open sets with $K \subseteq \bigcup_{i \in I} U_i$.

(ii) A **subcover** is a subfamily that still covers K. It is **finite** if it has finitely many members.

(iii) K is **compact** if *every* open cover of K has a finite subcover:

$$K \subseteq \bigcup_{i \in I} U_i \implies K \subseteq U_{i_1} \cup \cdots \cup U_{i_n} \quad \text{for some } i_1, \ldots, i_n \in I.$$

Warning (The most common misreading). "K has a finite open cover" is always true, since $\{X\}$ is one. The definition requires that *every* open cover can be reduced to finitely many sets. To prove compactness, handle an *arbitrary* cover. To disprove it, exhibit *one* cover with no finite subcover.

Definition 5.2 (Sequential compactness, total boundedness)**.**

(i) X is **sequentially compact** if every sequence in X has a subsequence converging to a point *of* X.

(ii) X is **totally bounded** if for every $\varepsilon > 0$ there are finitely many points $x_1, \ldots, x_n$ (an **ε-net**) with

$$X = B(x_1, \varepsilon) \cup \cdots \cup B(x_n, \varepsilon).$$

Totally bounded $\Rightarrow$ bounded, but not conversely. An infinite discrete space has diam $= 1$, but with $\varepsilon = \frac{1}{2}$ each ball is a single point.

5.2 The main equivalence

Theorem 5.3. *For a metric space X, the following are equivalent:*

(1) X is compact;

(2) X is sequentially compact;

(3) X is complete and totally bounded.

Sketch of proof. (1)⇒(2): Suppose (x_n) has no convergent subsequence. Then each $x \in X$ has a ball $B(x, r_x)$ containing x_n for only finitely many n. Finitely many of these balls cover X, so only finitely many indices n exist, a contradiction.

(2)⇒(3): A Cauchy sequence has a convergent subsequence, so it converges (Proposition 4.2). If X were not totally bounded for some ε, we could inductively choose $x_{n+1} \notin \bigcup_{k \le n} B(x_k, \varepsilon)$. Then $d(x_m, x_n) \ge \varepsilon$ for all $m \neq n$, so no subsequence is Cauchy.

(3)⇒(2): Cover X by finitely many balls of radius 1. One contains infinitely many terms. Repeat inside that with radius $\frac{1}{2}, \frac{1}{3}, \dots$ and take a diagonal subsequence. It is Cauchy, so it converges by completeness.

(2)⇒(1): Use the Lebesgue number δ of the cover (Lemma 5.11) and a finite $\frac{\delta}{3}$-net. Each ball of the net lies inside a single member of the cover. □

5.3 Heine–Borel and its limits

Theorem 5.4 (Compact ⇒ closed and bounded)**.** *A compact subset K of a metric space X is closed and bounded.*

Proof. Bounded: $\{B(x_0, n)\}_{n \in \mathbb{N}}$ is an open cover of K, and a finite subcover gives $K \subseteq B(x_0, N)$. *Closed:* let $k_n \in K$ with $k_n \to x$. Some subsequence converges to a point of K. That subsequence also converges to x, so by uniqueness of limits $x \in K$. □

Theorem 5.5 (Heine–Borel)**.** *A subset of $(\mathbb{R}^n, d_2)$ is compact if and only if it is closed and bounded.*

Proof. (⇒) Theorem 5.4. (⇐) A sequence in a bounded set has a convergent subsequence by Bolzano–Weierstrass, applied one coordinate at a time. Its limit lies in the set because the set is closed. So the set is sequentially compact. □

Warning (Closed and bounded ⇏ compact in general).

(a) An infinite discrete space is closed and bounded, but the cover $\{\{x\}\}_{x \in X}$ has no finite subcover.

(b) In the closed unit ball of ℓ^2, the vectors $e_n = (0, \dots, 0, 1, 0, \dots)$ satisfy $d(e_n, e_m) = \sqrt{2}$ for $n \neq m$, so no subsequence is Cauchy.

(c) In the closed unit ball of $(C[0,1], d_\infty)$, $f_n(t) = t^n$ has no uniformly convergent subsequence, because its pointwise limit is discontinuous.

(d) $X = (0, 1]$ is closed in itself and bounded, but $(\frac{1}{n})$ has no subsequence converging in X.

In general, the correct replacement for "closed and bounded" is *complete and totally bounded*.

5.4 Key theorems

Theorem 5.6. *A closed subset of a compact space is compact.*

Proof. Let $F \subseteq K$ be closed and $\{U_i\}$ an open cover of F. Then $\{U_i\} \cup \{X \setminus F\}$ is an open cover of K. Take a finite subcover and discard $X \setminus F$; the remaining sets cover F. □

Theorem 5.7 (Continuous image of a compact set)**.** *If $f\colon X \to Y$ is continuous and $K \subseteq X$ is compact, then $f(K)$ is compact.*

Proof. Let $\{V_i\}$ be an open cover of $f(K)$. Each $f^{-1}(V_i)$ is open (Theorem 3.2), and together they cover K. Choose a finite subcover $f^{-1}(V_1), \dots, f^{-1}(V_n)$. Then $f(K) \subseteq V_1 \cup \cdots \cup V_n$. □

Corollary 5.8 (Extreme Value Theorem)**.** *If $K \neq \emptyset$ is compact and $f\colon K \to \mathbb{R}$ is continuous, then f is bounded and attains its bounds: there exist $a, b \in K$ with*

$$f(a) = \inf_{x \in K} f(x), \qquad f(b) = \sup_{x \in K} f(x).$$

Proof. $f(K)$ is compact, hence closed and bounded in $\mathbb{R}$, so it contains its supremum and infimum. □

Without compactness this fails: $f(x) = x$ on $(0, 1)$ attains neither bound, and $1/x$ on $(0, 1]$ is unbounded.

Theorem 5.9 (Heine–Cantor)**.** *A continuous map from a compact metric space to a metric space is uniformly continuous.*

Proof. If not, by Proposition 3.7 there are $\varepsilon > 0$ and x_n, y_n with $d(x_n, y_n) < \frac{1}{n}$ but $d\big(f(x_n), f(y_n)\big) \geq \varepsilon$. By sequential compactness, $x_{n_k} \to x$ for some subsequence. Then $y_{n_k} \to x$ as well, since $d(y_{n_k}, x) \leq \frac{1}{n_k} + d(x_{n_k}, x)$. By continuity, $f(x_{n_k})$ and $f(y_{n_k})$ both converge to $f(x)$, so their distance tends to 0, a contradiction. □

Theorem 5.10. *Let $f\colon X \to Y$ be a continuous bijection, with X compact and Y Hausdorff (for example, a metric space). Then f is a homeomorphism.*

Proof. We must show f^{-1} is continuous, that is, $(f^{-1})^{-1}(C) = f(C)$ is closed for each closed $C \subseteq X$. C is compact (Theorem 5.6), so $f(C)$ is compact (Theorem 5.7), and compact subsets of Hausdorff spaces are closed. □

Lemma 5.11 (Lebesgue Number Lemma)**.** *If X is a compact metric space and $\mathcal{U}$ is an open cover of X, there is $\delta > 0$ (a **Lebesgue number**) such that every subset of X of diameter less than δ lies in a single member of $\mathcal{U}$.*

Theorem 5.12 (Finite intersection property)**.** *A family of sets has the **finite intersection property** (FIP) if every finite subfamily has non-empty intersection. X is compact $\iff$ every family of closed sets with the FIP has non-empty intersection. In particular, nested non-empty compact sets $K_1 \supseteq K_2 \supseteq \cdots$ satisfy $\bigcap_n K_n \neq \emptyset$.*

Proposition 5.13 (Further facts)**.**

(i) Finite unions of compact sets are compact.

(ii) A compact metric space is complete, totally bounded, bounded and separable.

(iii) $X \times Y$ *is compact* $\iff$ X *and* Y *are compact (Tychonoff's theorem extends this to arbitrary products).*

(iv) If K *is compact,* F *is closed and* $K \cap F = \emptyset$*, then* $d(K, F) > 0$*. This fails for two closed sets: the* x*-axis and* $\{\,(x, y) \,:\, y = 1/x,\ x > 0\,\}$ *are disjoint but at distance* 0*.*

(v) Compactness is a topological property.

Example 5.14 (Compactness in topological spaces)**.** Finite spaces, indiscrete spaces and cofinite spaces are compact. For a cofinite space, one member of any open cover misses only finitely many points, and finitely many more members cover those. An infinite discrete space is not compact.

5.5 Exercises

Exercise 5.1. [Easy] Show directly from the definition that $(0, 1)$ is not compact.

Solution. The sets $U_n = (\frac{1}{n}, 1)$, $n \geq 2$, cover $(0, 1)$. A finite subfamily has union $(\frac{1}{N}, 1)$, where N is its largest index, and this misses $\frac{1}{2N}$. □

Exercise 5.2. [Easy] Which of these are compact? (a) $[0, 1] \cup [2, 3]$; (b) $\{0\} \cup \left\{\, \frac{1}{n} \,:\, n \in \mathbb{N} \,\right\}$; (c) $\mathbb{Z}$; (d) $S^1 = \{\,(x, y) \,:\, x^2 + y^2 = 1\,\}$; (e) $\mathbb{Q} \cap [0, 1]$.

Solution. (a) Yes: closed and bounded. (b) Yes: closed, since it contains its only limit point 0, and bounded. (c) No: unbounded. (d) Yes: closed (a preimage of $\{1\}$) and bounded. (e) No: not closed, since $\frac{\sqrt{2}}{2}$ is a limit of points in the set but not in it. □

Exercise 5.3. [Medium] Prove from the open-cover definition that $K = \{0\} \cup \left\{\, \frac{1}{n} \,:\, n \in \mathbb{N} \,\right\}$ is compact.

Solution. Let $\mathcal{U}$ be an open cover. Some $U_0 \in \mathcal{U}$ contains 0, and hence contains $(-\varepsilon, \varepsilon)$ for some $\varepsilon > 0$, so it contains $\frac{1}{n}$ for all $n > \frac{1}{\varepsilon}$. The finitely many remaining points $1, \frac{1}{2}, \ldots, \frac{1}{N}$ each lie in some member of $\mathcal{U}$. That gives at most $N + 1$ sets in total. The limit point swallows the tail. □

Exercise 5.4. [Medium] Let K be compact and $f \colon K \to \mathbb{R}$ continuous with $f > 0$. Show $\inf_K f > 0$.

Solution. By Corollary 5.8, $\inf_K f = f(x_0)$ for some $x_0 \in K$, and $f(x_0) > 0$. Compare $f(x) = x$ on $(0, 1]$, where the infimum 0 is not attained. □

Exercise 5.5. [Hard] Let (X, d) be compact and $f \colon X \to X$ satisfy $d(f(x), f(y)) < d(x, y)$ for all $x \neq y$. Prove that f has a unique fixed point. Compare with $x + \frac{1}{x}$ on $[1, \infty)$ in Chapter 4.

Solution. $g(x) = d(x, f(x))$ is continuous on the compact space X, so it attains its minimum at some x_0. If $f(x_0) \neq x_0$, then

$$g(f(x_0)) = d(f(x_0), f(f(x_0))) < d(x_0, f(x_0)) = g(x_0),$$

contradicting minimality. So $f(x_0) = x_0$. Uniqueness: two distinct fixed points $x \neq y$ would give $d(x, y) < d(x, y)$. Compactness replaces the uniform constant $k < 1$. □

Exercise 5.6. [Hard] Show that an isometry $f\colon X \to X$ of a compact metric space is surjective.

Solution. Suppose $a \notin f(X)$. $f(X)$ is compact, hence closed, so $\varepsilon = d(a, f(X)) > 0$. Let $x_n = f^n(a)$. For $m > n$, since f^n preserves distances,

$$d(x_n, x_m) = d\big(a, f^{m-n}(a)\big) \geq \varepsilon,$$

because $f^{m-n}(a) \in f(X)$. So (x_n) has no Cauchy subsequence, contradicting sequential compactness. □

Appendix A

Summary of Chapters 0–2

These are the facts from the earlier chapters that Chapters 3–5 rely on. References such as "Theorem 2.9" point to the full Chapters 0–5 notes.

Metric (Definition 1.1). $d\colon X \times X \to \mathbb{R}$ with (M1) $d(x,y) \geq 0$ and $d(x,y) = 0 \iff x = y$; (M2) $d(x,y) = d(y,x)$; (M3) $d(x,z) \leq d(x,y) + d(y,z)$.

Standard examples. $\mathbb{R}$ with $|x-y|$. $\mathbb{R}^n$ with d_1, d_2, d_∞, where $d_\infty \leq d_2 \leq d_1 \leq n d_\infty$. The discrete metric. $C[a,b]$ with $d_\infty(f,g) = \sup|f-g|$ or $d_1(f,g) = \int |f-g|$. Normed spaces with $d(u,v) = \|u-v\|$. $f_n(t) = t^n$ has $d_1(f_n, 0) \to 0$ but $d_\infty(f_n, 0) = 1$.

Reverse triangle inequality. $|d(x,z) - d(y,z)| \leq d(x,y)$.

Balls. $B(x,r) = \{\, y \;:\; d(x,y) < r \,\}$ is open: for $y \in B(x,r)$, the ball $B\big(y, r - d(x,y)\big)$ lies inside it. $\overline{B}(x,r)$ is closed.

Open and closed sets. U is open if $\forall x \in U\ \exists r > 0 :\ B(x,r) \subseteq U$. F is closed if $X \setminus F$ is open. Arbitrary unions and finite intersections of open sets are open. A set can be both open and closed, or neither.

Closure (Theorem 2.9). The following are equivalent: $x \in \operatorname{cl} A$; every ball around x meets A; $x = \lim a_n$ with $a_n \in A$; $d(x,A) = 0$. F is closed $\iff$ F contains the limits of all its convergent sequences.

Interior, boundary, dense. $\operatorname{int} A$ is the largest open subset of A, and $\partial A = \operatorname{cl} A \setminus \operatorname{int} A$. A is dense if $\operatorname{cl} A = X$. X is separable if it has a countable dense subset.

Convergence. $x_n \to x \iff d(x_n, x) \to 0$. Limits are unique. In $\mathbb{R}^n$, convergence is coordinatewise. In $(C[a,b], d_\infty)$, it is uniform convergence. In a discrete space, a convergent sequence is eventually constant.

Topology (Definition 2.16). A family τ containing $\emptyset$ and X, closed under arbitrary unions and finite intersections. Examples: discrete, indiscrete, cofinite. **Hausdorff**: distinct points have disjoint open neighbourhoods. Every metric space is Hausdorff, so a non-Hausdorff space is not metrizable.

Subspace. $U \subseteq Y$ is open in $Y \iff U = V \cap Y$ with V open in X. For example, $[0,1)$ is open in $[0,2]$.

Preimages commute with unions, intersections and complements, but images do not.

Appendix B

Connectedness

Some courses put connectedness in Chapter 5 or 6. It is short and often appears in questions about homeomorphisms.

Definition B.1.

(i) X is **disconnected** if $X = U \cup V$ with U, V open, disjoint and non-empty (a **separation**). Otherwise X is **connected**. Equivalently, the only clopen subsets are $\emptyset$ and X, or every continuous $f\colon X \to \{0,1\}$ is constant.

(ii) X is **path-connected** if for all $x, y \in X$ there is a continuous $\gamma\colon [0,1] \to X$ with $\gamma(0) = x$ and $\gamma(1) = y$.

(iii) A **component** is a maximal connected subset.

Theorem B.2.

(i) The connected subsets of $\mathbb{R}$ are exactly the intervals.

*(ii) Continuous images of connected spaces are connected. This gives the **Intermediate Value Theorem**.*

*(iii) Path-connected $\Rightarrow$ connected. The converse fails for the **topologist's sine curve***

$$S = \left\{ \left(x, \sin \tfrac{1}{x}\right) \;:\; x > 0 \right\} \cup (\{0\} \times [-1,1]).$$

(iv) Open connected subsets of $\mathbb{R}^n$ are path-connected.

(v) The closure of a connected set is connected. A union of connected sets with a common point is connected.

(vi) $\mathbb{Q}$ is totally disconnected: $\left\{ q \;:\; q < \sqrt{2} \right\}$ and $\left\{ q \;:\; q > \sqrt{2} \right\}$ separate it.

Exam tip (Using connectedness to show $X \ncong Y$). If $f\colon X \to Y$ is a homeomorphism, then $X \setminus \{p\} \cong Y \setminus \{f(p)\}$. Removing any point disconnects $\mathbb{R}$ but not $\mathbb{R}^2$, so $\mathbb{R} \ncong \mathbb{R}^2$. Similarly $[0,1) \ncong (0,1)$ and $S^1 \ncong [0,1]$.

Appendix C

Proof-Template Toolkit

Most exam questions follow one of these patterns. Learn the first move for each.

To prove...	**Standard first move**
d is a metric	Check (M1), (M2), (M3). For (M3), add and subtract, or use an increasing subadditive φ.
d is not a metric	One numerical counterexample to one axiom.
U is open	"Let $x \in U$. Take $r = \dots$", then show $B(x, r) \subseteq U$. Or write $U = f^{-1}(\text{open})$ with f continuous.
F is closed	Show the complement is open; or $F = f^{-1}(\text{closed})$; or $x_n \in F,\ x_n \to x \Rightarrow x \in F$.
A set is not open	One point of the set where every ball leaves the set.
A set is not closed	A sequence in the set converging to a point outside it.
$x \in \operatorname{cl} A$	A sequence in A converging to x, or show every ball around x meets A.
f is continuous	ε–δ; build it from known continuous maps; or show it is Lipschitz.
f is not continuous	One sequence $x_n \to x$ with $f(x_n) \not\to f(x)$.
f is not uniformly continuous	$d(x_n, y_n) \to 0$ but $d\big(f(x_n), f(y_n)\big) \geq \varepsilon$.
$X \cong Y$	An explicit bijection, continuous in both directions; or a continuous bijection from a compact space to a Hausdorff space.
$X \ncong Y$	A topological property that differs: compactness, connectedness, components after removing a point, Hausdorff, separability.
(x_n) is Cauchy	Bound $d(x_m, x_n)$ for $m > n$ by a quantity depending only on n that tends to 0. A geometric series often works.
X is complete	A closed subset of a complete space, or the 3-step template: candidate limit, convergence in the metric, membership.
X is not complete	One Cauchy sequence whose limit is missing.
A unique solution exists	The Contraction Mapping Theorem: complete space, $f(X) \subseteq X$, constant $k < 1$.
K is compact	In $\mathbb{R}^n$: closed and bounded. Otherwise: sequential compactness, a closed subset of a compact set, a continuous image of a compact set, or an arbitrary open cover.
K is not compact	One open cover with no finite subcover; one sequence with no convergent subsequence; or show it is not closed, or not bounded.
(X, τ) is not metrizable	Show it is not Hausdorff.

Appendix D

Counterexample Bank

False statement	**Counterexample**
Not open $\Rightarrow$ closed	$[0,1)$, or $\mathbb{Q}$, in $\mathbb{R}$
Infinite intersections of open sets are open	$\bigcap_n (-\frac{1}{n}, \frac{1}{n}) = \{0\}$
Infinite unions of closed sets are closed	$\bigcup_n [\frac{1}{n}, 1] = (0,1]$
$\operatorname{cl} B(x,r) = \overline{B}(x,r)$	discrete metric with $r = 1$
$\operatorname{cl}(A \cap B) = \operatorname{cl} A \cap \operatorname{cl} B$	$A = \mathbb{Q}$, $B = \mathbb{R} \setminus \mathbb{Q}$
Limits are unique in every topological space	indiscrete or cofinite topology
Continuous images of open or closed sets are open or closed	x^2 on $(-1,1)$; arctan on $\mathbb{R}$
Continuous $\Rightarrow$ uniformly continuous	x^2 on $\mathbb{R}$; $1/x$ on $(0,1)$
Uniformly continuous $\Rightarrow$ Lipschitz	$\sqrt{x}$ on $[0,1]$
A continuous bijection is a homeomorphism	$[0, 2\pi) \to S^1$; $\mathrm{id}\colon (\mathbb{R}, \mathrm{disc}) \to (\mathbb{R}, \lvert\cdot\rvert)$
Boundedness and completeness are topological	$\mathbb{R} \cong (0,1)$
Topologically equivalent $\Rightarrow$ Lipschitz equivalent	d and $d/(1+d)$ on $\mathbb{R}$
d_1 and d_∞ on $C[0,1]$ are equivalent	t^n: $d_1 \to 0$ but $d_\infty = 1$
$x_{n+1} - x_n \to 0 \Rightarrow$ Cauchy	$\sum_{k \le n} \frac{1}{k}$; $\sqrt{n}$
Every Cauchy sequence converges	$\frac{1}{n}$ in $(0,1)$; approximations to $\sqrt{2}$ in $\mathbb{Q}$
$d(fx, fy) < d(x,y) \Rightarrow$ fixed point	$x + \frac{1}{x}$ on $[1, \infty)$ (true if X is compact)
A contraction on any space has a fixed point	$x/2$ on $(0,1]$
Nested non-empty closed sets intersect	$[n, \infty)$ in $\mathbb{R}$
Closed and bounded $\Rightarrow$ compact	infinite discrete space; unit ball of ℓ^2; $(0,1]$ in itself
Bounded $\Rightarrow$ totally bounded	infinite discrete space
Disjoint closed sets are a positive distance apart	x-axis and $y = 1/x$
Connected $\Rightarrow$ path-connected	topologist's sine curve

Appendix E

Glossary

Banach space
A complete normed vector space.

Bounded
Contained in some ball; equivalently, $\operatorname{diam} A < \infty$.

Cauchy sequence
$\forall \varepsilon\ \exists N\ \forall m, n \geq N :\ d(x_m, x_n) < \varepsilon$.

Compact
Every open cover has a finite subcover.

Complete
Every Cauchy sequence converges in the space.

Completion
A complete space containing X isometrically as a dense subset.

Connected
Not a union of two disjoint non-empty open sets.

Continuous
Preimages of open sets are open; equivalently, the ε–δ or sequential conditions.

Contraction
$d(f(x), f(y)) \leq k\, d(x, y)$ for a fixed $k < 1$.

Dense
$\operatorname{cl} A = X$; equivalently, A meets every non-empty open set.

Diameter
$\operatorname{diam} A = \sup \{\, d(a, b) \ :\ a, b \in A \,\}$.

Distance to a set
$d(x, A) = \inf_{a \in A} d(x, a)$; it is $0 \iff x \in \operatorname{cl} A$.

Finite intersection property
Every finite subfamily has non-empty intersection.

Fixed point
A point with $f(x) = x$.

Hausdorff
Distinct points have disjoint open neighbourhoods.

Homeomorphism
A continuous bijection with a continuous inverse.

Isolated point
A point $a \in A$ with $B(a, r) \cap A = \{a\}$ for some $r > 0$.

Isometry
A distance-preserving map.

Lebesgue number
A δ such that every set of diameter $< \delta$ lies in one member of the cover.

Lipschitz
$d(f(x), f(y)) \le L\, d(x, y)$.

Nowhere dense
$\operatorname{int}(\operatorname{cl} A) = \emptyset$.

Open cover
A family of open sets whose union contains the set.

Path-connected
Any two points are joined by a continuous path.

Separable
Has a countable dense subset.

Sequentially compact
Every sequence has a subsequence converging in the space.

Topological property
A property preserved by homeomorphisms.

Totally bounded
For every $\varepsilon > 0$, finitely many ε-balls cover the space.

Uniformly continuous
The ε–δ condition with δ independent of the point.

Appendix F

Recap Checklist (5:35–5:45)

Close the notes. On a blank page, write as many of these as you can from memory. Tick off what you got and circle what you missed. The circled items are where to start next session.

Definitions to state exactly

- ☐ Continuity (ε–δ and open-set versions); uniform continuity; Lipschitz
- ☐ Homeomorphism; topological property
- ☐ Cauchy sequence; complete; contraction
- ☐ Compact (open covers); sequentially compact; totally bounded

Theorems to state, with a proof sketch

- ☐ The four equivalent forms of continuity
- ☐ Closed subsets of complete spaces are complete; complete subspaces are closed
- ☐ $(C[a,b], d_\infty)$ is complete (the $\varepsilon/3$ argument)
- ☐ Contraction Mapping Theorem: all five steps
- ☐ Compact $\iff$ sequentially compact $\iff$ complete and totally bounded
- ☐ Heine–Borel, and a counterexample outside $\mathbb{R}^n$
- ☐ The continuous image of a compact set is compact, hence the Extreme Value Theorem
- ☐ Heine–Cantor; a continuous bijection from a compact space to a Hausdorff space is a homeomorphism

Counterexamples to recall instantly

- ☐ A set that is neither open nor closed
- ☐ A continuous function that is not uniformly continuous
- ☐ A continuous bijection that is not a homeomorphism
- ☐ An incomplete space, with its Cauchy sequence
- ☐ A closed, bounded, non-compact set

> *Exam tip* (Next up). Break 5:45–6:00, then **linear systems 6:00–6:50** (separate notes), then wrap-up at 6:50 and a hard stop at **7:00 pm**.